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ORIGINAL ARTICLE

# Methodology reporting improved over time in 176,469 randomized controlled trials

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**Abstract**

**Objectives:** To describe randomized controlled trial (RCT) methodology reporting over time.

**Study Design and Setting:** We used a deep learning-based sentence classification model based on the Consolidated Standards of Reporting Trials (CONSORT) statement, considered minimum requirements for reporting RCTs. We included 176,469 RCT reports published between 1966 and 2018. We analyzed the reporting trends over 5-year time periods, grouping trials from 1966 to 1990 in a single stratum. We also explored the effect of journal impact factor (JIF) and medical discipline.

**Results:** Population, Intervention, Comparator, Outcome (PICO) items were commonly reported during each period, and reporting increased over time (e.g., interventions: 79.1% during 1966−1990 to 87.5% during 2010−2018). Reporting of some methods information has increased, although there is room for improvement (e.g., sequence generation: 10.8−41.8%). Some items are reported infrequently (e.g., allocation concealment: 5.1−19.3%). The number of items reported and JIF are weakly correlated (Pearson's $r$ (162,702) = 0.16, $P < 0.001$). The differences in the proportion of items reported between disciplines are small (<10%).

**Conclusion:** Our analysis provides large-scale quantitative support for the hypothesis that RCT methodology reporting has improved over time. Extending these models to all CONSORT items could facilitate compliance checking during manuscript authoring and peer review, and support metaresearch. 

*Keywords:* Randomized controlled trials; Reporting guidelines; CONSORT; Text mining; Machine learning; Meta-research

## 1. Introduction

Incomplete reporting of study methods in biomedical publications hinders efforts to evaluate and apply research findings [1,2]. While complete reporting alone does not guarantee that research is rigorous or that research results can be applied in various settings, complete reporting is essential for assessing these criteria [3]. Reporting guidelines have been developed to improve the completeness of reporting, and they have been endorsed by many high-impact medical journals [4]. However, adherence to reporting guidelines remains suboptimal [5].

Randomized controlled trials (RCTs) are a cornerstone of evidence-based medicine [6]. They are used to make regulatory decisions, and they provide evidence for clinical guidelines and practice. Rigorous design and conduct, combined with complete and accurate reporting, are essential for fully leveraging the strengths of RCTs. Reporting information about trial methods (e.g., sample size calculation, randomization, and masking) and results (e.g., effect estimates) is

Data availability: RCT publication metadata is available at https://github.com/wmotte/RCTQuality/tree/main/data. Article-level text mining model predictions that were used in the analyses are available at https://github.com/ScienceNLP-Lab/RCT-Transparency/tree/main/ReportingQualityAnalysis.

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